THE TOM KLOOTWIJK MANIFOLD SYSTEM

A rigorous definition, claim audit, and executable verification

Formal specification TKM 1.0

Prepared for	Tom Klootwijk
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Source basis: UGTS-KC 2.0; two requester-supplied conversational PDFs; KLB SeedChain GPU 0.3.0; standard mathematical and computer-science literature.

Document boundary and reading rule

Non-negotiable epistemic boundary

This report does three different things and labels them separately:

1. it **formalizes** a coherent object under the requested name "Tom Klootwijk Manifold System";
2. it **checks** mathematical claims by proof, counterexample, or exact calculation;
3. it **reproduces** the CPU-verifiable portion of the supplied KLB package and audits the bundled GPU evidence.

It does not convert a requester-supplied name into a recognized theorem, and it does not turn a benchmark log into a universal hardware law. A statement is promoted only to the level supported by its evidence.

The strongest conceptual source, UGTS-KC 2.0, already imposes the correct reading rule: geometry supplies typed patterns and fields; topology supplies explicit gluing and route classes; kinematics supplies state-at-time and moving frames; dynamics supplies bounded flows; and discrete change remains controlled by compatibility-gated events with lineage. This report retains that architecture and specializes it into a named system.

The two conversational PDFs contain useful motifs but also contain mutually contradictory and technically unsupported AI-generated passages. In particular, one passage correctly computes that 943 and 937 differ in two binary positions, while later passages claim that zero-based indexing or a fractal fold makes the same transition a one-bit mutation. Those cannot both be true under the same encoding. This report resolves such conflicts by explicit definitions and tests rather than by selecting the most flattering passage.

Label	Meaning	Promotion rule
VERIFIED	Verified or reproduced	Established by proof, exact arithmetic, hash/integrity check, or a clean reproduction in this environment.
CONDITIONAL	Conditionally valid	Correct only when its stated hypotheses, error budget, and application boundary hold.
REJECTED AS STATED	Rejected as stated	False, undefined, or overbroad in the form found in the source material.
SOURCE-ASSERTED	Source-asserted only	Supported by supplied logs/reports but not independently rerun here, usually because the required GPU or external reference was unavailable.
METAPHOR / EXCLUDED	Metaphor / excluded	Symbolic analogy that is not used as a mathematical or physical premise.

Executive verdict

The corrected object

The **Tom Klootwijk Manifold System, version 1.0** is a typed query-and-event system built over a seven-dimensional smooth *ambient state manifold* X_7, together with a scalar field, a finite address complex, a bounded flow, verified events, lineage, and an error contract:

$$\mathbf{TKM}_1 = \big(X_7, g;\ Q;\ f;\ \mathcal{A}, \pi, \Gamma;\ \Phi;\ \mathcal{E};\ \mathcal{L};\ \varepsilon\big).$$

The object called the *Tom Klootwijk boundary* is $B_{\mathrm{TK}} = f^{-1}(0)$. If 0 is a regular value of $f : X_7 \to \mathbb{R}$, then B_{TK} is a smooth embedded **six-dimensional** submanifold. The seven-dimensional object is the ambient state manifold; one scalar equation normally removes one smooth dimension.

The formalization preserves four source ideas in defensible form:

- **Seven orthogonal directions:** seven mutually orthogonal axes are possible in a seven-dimensional inner-product space. They are not possible in a four-dimensional space.
- **SDF/implicit boundary:** $f = 0$ selects a zero-level set. It does not, by itself, make that set spherical, smooth, or an exact Euclidean signed-distance field.
- **Compact address and lineage:** a finite address grammar can identify routes, sheets, or reconstructible states compactly, but relabeling cannot change Hamming distance or information content.
- **Chronotemporal latch:** the defensible meaning is deterministic state reconstruction with checkpoint and lineage consistency, not time teleportation by a parity bit.

Verification in one page

Formal geometry	A coherent seven-dimensional ambient model and a regular six-dimensional boundary are defined.
Bit/address claim	$943 \oplus 937 = 6 = 0000000110_2$, so the Hamming distance is 2, not 1. Zero-based indexing and LSB-first display do not alter that invariant.
Sierpinski claim	The standard triangle uses three contraction maps and the natural alphabet $\{0, 1, 2\}$; two-bit storage may reserve a fourth code but does not make the geometry intrinsically base four.
Package integrity	All 82 files listed in the supplied SHA-256 manifest verified.
CPU build/tests	A clean GCC 14.2, C++20, CPU-only build completed; both test suites passed.
Application reproduction	The complete seven-day pass schedule was regenerated: 604,800 intervals, 19,353,600 candidates, 19,214,155 support/compatible survivors, 5,498,030 visible samples, and 717 events. The regenerated CSV was bit-for-bit identical to the bundled CSV.
GPU evidence	Bundled logs show a narrow direct-seed advantage over materialize-plus-query (0.93–2.50%), but a roughly 4.5x penalty versus resident dense lookup. These results were not independently rerun here.
Operational accuracy	Not established: no SGP4 comparison is supplied, and a stronger bundled full-horizon check reports a one-count CPU/GPU support mismatch.

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Chapter 1

Purpose, source corpus, and evidence hierarchy

1.1 Purpose

The requester asked for a comprehensive, formally stated definition of the "Tom Klootwijk manifold" and for verification against the supplied substrate documents and implementation package. The challenge is not merely to rename the earlier prose. The source corpus mixes four different layers:

1. a mature typed substrate architecture (UGTS-KC 2.0);
2. informal signed-distance and symbolic analogies;
3. a conversational namesake proposal with internal contradictions;
4. an executable C++/CUDA package that deliberately narrows speculative motifs into testable records, reconstruction, support predicates, guards, and events.

A formal definition must therefore separate the smooth manifold, the zero-level boundary, the discrete address complex, and the executable event system. Calling all four the same "manifold" would hide the most important distinctions.

1.2 Source corpus

Table 1.1: *Requester-supplied source corpus and treatment.*

ID	Source	SHA-256	Treatment
S1	*UGTS-KC 2.0: Pattern, Kinematic Calculus and Dynamics Expansion* (23 pages)	`a29dd995...e878399`	Primary conceptual basis. Its typed field, query-first event, topology, kinematic, validation, and claim boundaries are retained.
S2	*The Tom Klootwijk exact SDF notation...* (11 pages)	`b3ab8e1d...49b6aa3`	Conversational source. Basic SDF sign/zero semantics are retained; quantum and "chronotemporal" extrapolations are audited rather than assumed.
S3	*THE Tom Klootwijk... manifold and not the Klootwijk family folding* (25 pages)	`5004ef70...6b97d2`	Namesake and notation source. The proposed name and mnemonic are preserved; contradictory mathematical claims are resolved by proof and explicit definitions.
S4	*KLB SeedChain GPU 0.3.0* ZIP	`c3e00e94...4d1ad4`	Executable evidence. Integrity, CPU build, tests, CLI outputs, and seven-day schedule were independently reproduced. Bundled GPU logs were audited but not rerun.

The full 64-hex-digit hashes are reproduced in section .0. The names, identifier, and date attached to the corpus remain requester-supplied provenance. S1 and S4 explicitly state that these fields do not independently prove identity or authorship.

1.3 Evidence ladder

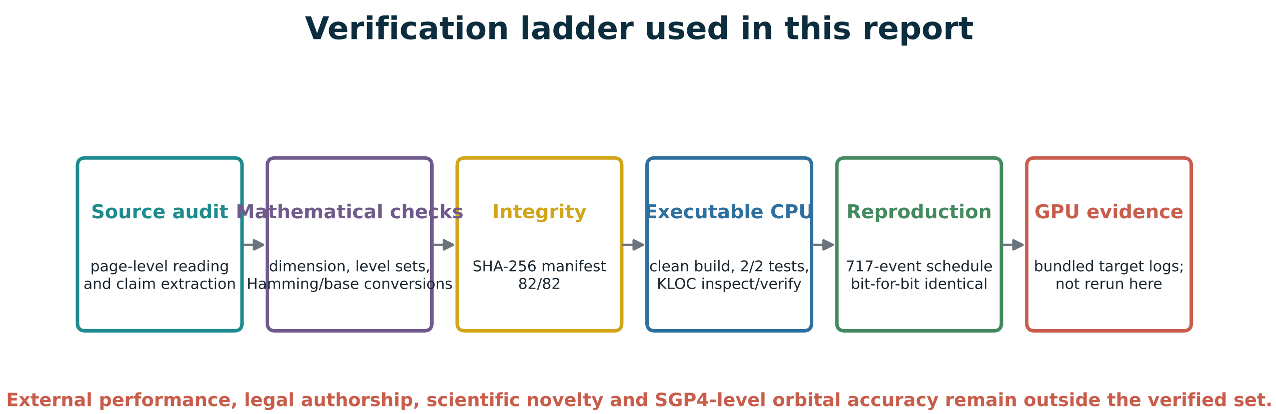


Figure 1.1: *The verification ladder used in this report. Evidence is not promoted above the rung actually reached.*

The rungs are intentionally different. A dimensional theorem can be proved without code. A package hash can be verified without proving the physics. A CPU reproduction can establish deterministic implementation behavior while leaving GPU timing and orbital accuracy open. This prevents a common category error: treating compact notation, executable consistency, hardware performance, and physical truth as interchangeable.

1.4 Why the conversational sources require reconciliation

S2 and S3 repeatedly include the warning that AI responses may contain mistakes. More importantly, the text itself demonstrates the problem. S3 first gives the correct binary calculation and says the transition 943 → 937 needs two bit changes. It later claims a zero-indexed fractal interpretation turns the same transition into one bit, then later gives a formal critique that again says the Hamming distance remains two, and finally returns to the one-bit claim. No single formal system can adopt all of these statements simultaneously.

The appropriate engineering treatment is the one already modeled by S1 and S4: preserve motifs only after assigning them types, domains, invariants, and validation contracts. Under that approach, “LSB,” “SDF 0,” “fractal,” “7D,” and “parity” cease to be free-floating metaphors and become separate fields in a schema.

Mathematical foundation

2.1 Manifolds and the dimensional correction

A smooth n-manifold is a Hausdorff, second-countable space locally diffeomorphic to open subsets of $\mathbb{R}^n$. The word *dimension* refers to the number of independent local coordinates in the manifold itself, not to the length of a displayed tuple unless that tuple is explicitly declared to be a coordinate chart.

Orthogonality bound

In an n-dimensional inner-product space, any set of nonzero mutually orthogonal vectors is linearly independent, so it contains at most n vectors. Consequently, seven mutually orthogonal axes are possible in $\mathbb{R}^7$ and impossible in $\mathbb{R}^4$.

Proof. If $v_1, \dots, v_k$ are nonzero and mutually orthogonal and $\sum_i a_i v_i = 0$, taking the inner product with v_j gives $a_j \left\|v_j\right\|^2 = 0$, hence every $a_j = 0$. Thus the vectors are linearly independent, so $k \le n$. □

This verifies the defensible part of the "seven perpendicular lines" motif: it works as a statement about a seven-dimensional inner-product space. The tuple $(4, 4, 4, 4)$ has four entries and does not create seven axes.

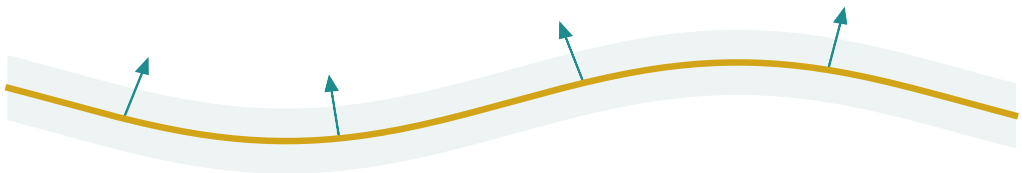
2.2 A scalar zero set normally loses one dimension

Let X_7 be a smooth seven-dimensional manifold and let $f : X_7 \longrightarrow \mathbb{R}$ be smooth. If 0 is a regular value, meaning $df_x \neq 0$ at every $x \in f^{-1}(0)$, then the regular level-set theorem gives

$$B_{\mathrm{TK}} = f^{-1}(0), \qquad \dim B_{\mathrm{TK}} = 7 - 1 = 6.$$

Regular level-set principle

A scalar equation removes one smooth dimension when df != 0 on the zero set.



2D analogy: a regular zero set is a 1D curve.

Therefore: f: X7 -> R gives a 6D boundary B = f^(-1)(0).

To make the zero set itself 7D, use an 8D ambient manifold.

Figure 2.1: *Regular level-set principle. A single independent scalar constraint removes one smooth dimension.*

This is the central dimensional correction. The report therefore reserves the name *Tom Klootwijk ambient manifold* for X_7 and the name *Tom Klootwijk boundary manifold* for the regular zero set B_{TK}. To obtain a seven-dimensional zero set from one scalar equation, the ambient manifold would need dimension eight.

If $df = 0$ somewhere on the zero set, singularities can occur and the zero set need not be a manifold there. A formal definition must either exclude those points, stratify them, or return a failure/status object rather than silently calling the whole set smooth.

2.3 Exact signed distance versus generic implicit field

Given a closed set K in a Riemannian manifold (X, g), the unsigned distance is

$$d_K(x) = \inf_{y \in K} d_g(x, y).$$

When an inside/outside convention exists, a signed distance can be formed by assigning opposite signs on the two sides. An *exact* signed-distance field satisfies $|f(x)| = d_g(x, B)$ in its valid domain. Away from the medial axis and other nondifferentiable points, it obeys the eikonal property $\|\nabla f\|_g = 1$.

A generic implicit field only needs $f^{-1}(0)$ to represent a boundary. Multiplying it by a nonzero function, smoothing it, or combining it with other fields may preserve the zero set while destroying exact distance values. This distinction is explicit in UGTS-KC 2.0 and is essential here.

What “SDF 0” does and does not say

The condition $f(x) = 0$ says that x lies on the selected zero-level set. It does **not** by itself say that:

- f is an exact Euclidean signed-distance function;

- the zero set is a sphere or hypersphere;
- the zero set is smooth;
- a ray reaches it in one step;
- the field has no storage or computational cost.

Those properties require separate hypotheses.

2.4 The discrete address layer is not the smooth manifold

A finite grammar, IFS address, bit stream, route label, braid word, or cell complex is a discrete combinatorial object. It can index charts or supports on a manifold, but it does not become a smooth manifold merely because its symbols are interpreted geometrically.

For the standard Sierpinski triangle, one convenient IFS consists of three contractions F_0, F_1, F_2 of ratio $1/2$. Finite addresses are words

$$\mathcal{A}_d = \bigcup_{k=0}^{d} \{0, 1, 2\}^k,$$

and the similarity dimension s satisfies $3(1/2)^s = 1$, hence

$$s = \frac{\log 3}{\log 2} \approx 1.5849625007.$$

Two bits are enough to store three symbols, so an implementation may encode $0, 1, 2$ as $00, 01, 10$ and reserve 11. That storage convenience does not change the natural branching count from three to four.

2.5 Hamming distance is invariant under indexing conventions

For equal-length binary strings a, b, the Hamming distance is

$$d_H(a, b) = \#\{j : a_j \neq b_j\} = \operatorname{popcount}(a \oplus b).$$

Changing whether positions are numbered from zero or one does not change the bits. Reversing an agreed display order or applying the same coordinate permutation to both strings also preserves the count of unequal coordinates.

The supplied numbers are

$$943 = 1110101111_2 = 32233_4, \qquad 937 = 1110101001_2 = 32221_4.$$

Therefore

$$943 \oplus 937 = 6 = 0000000110_2, \qquad d_H(943, 937) = 2.$$

A simple parity bit can detect odd parity changes but does not by itself identify and correct an arbitrary erroneous bit. Hamming-style single-error correction requires a code with multiple parity checks. In no case does zero-based indexing turn the two unequal data positions above into one.

Chapter 3

Formal definition of the Tom Klootwijk Manifold System

3.1 The ambient manifold

Definition 1: seven-dimensional ambient state manifold

A **Tom Klootwijk ambient manifold** is a smooth seven-dimensional Riemannian manifold (X_7, g). One admissible reference model is

$$X_7 = \mathbb{R}^3_p \times \mathbb{R}_t \times \mathbb{S}^1_\phi \times \mathbb{R}_w \times \mathbb{R}_\rho.$$

Here p is position, t is time, ϕ is a cyclic phase, w is an unwrapped route/winding coordinate, and ρ is a profile or support coordinate. The particular interpretation is schema-bound; the dimension is not inferred from a slogan or tuple.

The model is illustrative rather than mandatory. Applications may choose any diffeomorphic or locally equivalent seven-coordinate model, including charted products with boundaries, provided chart transitions are explicit and compatible. Discrete fields such as sheet, branch, mode, and lineage are not forced into smooth coordinates.

3.2 The typed state bundle

Definition 2: state bundle

Let $Q \to X_7$ be a typed state bundle. A state record may include

$$q = (x, R, v, \omega, \kappa, \tau, m, s, b, \pi_{\text{policy}}, u, \ell, d),$$

where $x \in X_7$, $R \in SO(3)$ is orientation, v and ω are linear and angular velocity, κ and τ are curvature and torsion data, m is a dynamic mode, s is a sheet label, b is a branch label, u is uncertainty, ℓ is lineage, and d is optional dynamic state.

The total state space can have dimension greater than seven because the phrase "seven-dimensional" applies to the declared ambient base X_7, not to every attached measurement and control variable. This removes the ambiguity present in the conversational sources, where four-vector, seven-dimensional substrate, time, phase, and bit address were sometimes counted together and sometimes not.

3.3 Field and boundary contract

Definition 3: typed field and Tom Klootwijk boundary

A field instance is

$$\mathfrak{f} = (f, \theta, U, \mathsf{units}, \mathsf{kind}, \mathsf{derivatives}, \varepsilon_f),$$

where $f_\theta : U \subseteq X_7 \to \mathbb{R}$ is either an exact signed-distance field or a generic implicit field. Its zero boundary is

$$B_{\mathrm{TK}}(\theta) = \{x \in U : f_\theta(x) = 0\}.$$

If f_θ is smooth and 0 is regular, $B_{\mathrm{TK}}(\theta)$ is the six-dimensional **Tom Klootwijk boundary manifold**. Otherwise the query returns a singular/stratified status rather than falsely certifying a smooth manifold.

A field contract records units, domain, parameter bounds, derivative availability, solver family, residual, and event margin. This follows the UGTS-KC rule that an object is not just a label such as "spiral," "SDF," or "gyroid"; it is a versioned operator with declared validation limits.

3.4 Finite address complex and chart map

Definition 4: address complex

$\mathcal{A}$ is a finite or bounded-depth address language, and $\pi : \mathcal{A} \to C$ maps addresses to charts, cells, supports, or grammar instances. Each address has an explicit alphabet, bit encoding, depth bound, and invalid/reserved codes. Address equality is not substituted for geometric equality, and coordinate equality is not substituted for identity.

For a Sierpinski-style layer, the canonical alphabet is ternary. A quaternary storage representation is allowed only as an encoding contract with a specified meaning for the fourth code. Route identity may additionally store winding, covering lift, braid word, cell incidence, or chart transition data.

3.5 Finite grammar

Definition 5: finite typed grammar

Γ is a finite, versioned grammar that emits typed pattern, field, topology, and state operators. A minimal expression language is

$$\begin{aligned}\mathsf{Expr} ::= {} & \mathsf{Primitive}(\theta) \mid \mathsf{Transform}(F, \mathsf{Expr}) \mid \mathsf{Restrict}(D, \mathsf{Expr}) \\ & \mid \mathsf{Repeat}(P, N, \mathsf{Expr}) \mid \mathsf{Sweep}(P, C, \mathsf{framePolicy}) \mid \mathsf{FieldGuard}(f, c, \delta).\end{aligned}$$

Hard limits include grammar depth, symbol count, control points, period cells, derivative order, and address depth.

This permits compact procedural reconstruction without claiming that arbitrary scenes, arbitrary physics, or arbitrary histories are compressible by the same grammar.

3.6 Kinematic and dynamic flow

Definition 6: bounded state flow

Φ is a partial flow or bounded numerical evolution on Q:

$$\Phi : \mathcal{D} \subseteq \mathbb{R} \times Q \to Q, \qquad \Phi_t(q) = \mathsf{state\_at}(q, t).$$

Closed-form families may use $SE(2)/SE(3)$ exponentials, quaternions, screw interpolation, or analytic pattern laws. Other families use a declared integrator and error bound. Every query returns status, residual, and validity interval.

The expression-size cost of a closed-form state query may be independent of the number of omitted display frames, but this is not a universal $O(1)$ statement. The input size, grammar depth, precision, root solving, event count, and hardware execution all remain part of the complexity model.

3.7 Verified events and transition authority

Definition 7: verified event calculus

For event family j, let $S_j(q)$ be a support predicate, $C_j(q)$ a compatibility predicate, $g_j(q)$ a guard, σ_j a degeneracy classification, and T_j a transition/reset. A candidate at time t is verified only if

$$\begin{aligned} \mathsf{verified}_j(q,t) \iff{} & S_j(q_t) \wedge C_j(q_t) \wedge \left(\sigma_j \in \Sigma_j^{\text{accepted}}\right) \\ & \wedge\ \mathsf{confidence}_j \geq c_j^{\min} \wedge\ \mathsf{numericError}_j \leq \mathsf{eventMargin}_j. \end{aligned}$$

Only a verified event may commit $q^+ = T_j(q^-)$ and append a lineage record.

A self-intersection, zero of a periodic field, bit parity change, or topological crossing is therefore only a candidate. Sheet, phase, route class, mode, policy, uncertainty, and lineage still determine whether an interaction is authoritative.

3.8 Lineage, novelty, and validation

Definition 8: lineage and reconstructibility

$\mathcal{L}$ is an ordered log of verified transitions, topology patches, mode resets, external inputs, calibration changes, conflicts, and checkpoints. A compact seed/grammar representation is valid over horizon H only when

$$d_Q(\mathsf{reconstruct}(s, \Gamma, \mathcal{L}_{\leq t}, t),\ \Phi_t(q_0)) \leq \varepsilon(t) \quad \text{for every declared } t \in H.$$

Irreducible external novelty must be stored; it may not be invented by the reconstructor.

The validation object ε contains units, parameter bounds, numerical floors, error tolerances, event margins, determinism requirements, performance targets, and kill criteria. Hashes may check integrity, but non-cryptographic hashes do not prove identity or authorship.

3.9 The complete structured tuple

Tom Klootwijk Manifold System TKM 1.0

$$\boxed{\mathbf{TKM}_1 = \left(X_7, g;\ Q;\ f;\ \mathcal{A}, \pi, \Gamma;\ \Phi;\ \mathcal{E};\ \mathcal{L};\ \varepsilon\right)}$$

with the following canonical authority path:

$$\boxed{\begin{array}{c}\text{typed pattern/field and state} \longrightarrow \text{local support} \longrightarrow \text{compatibility} \\ \longrightarrow \text{guard classification} \longrightarrow \text{verified event} \\ \longrightarrow \text{transition/reset} \longrightarrow \text{lineage and novelty}\end{array}}$$

The smooth ambient manifold, its zero boundary, the discrete address complex, and the hardware realization are distinct layers connected by typed maps.

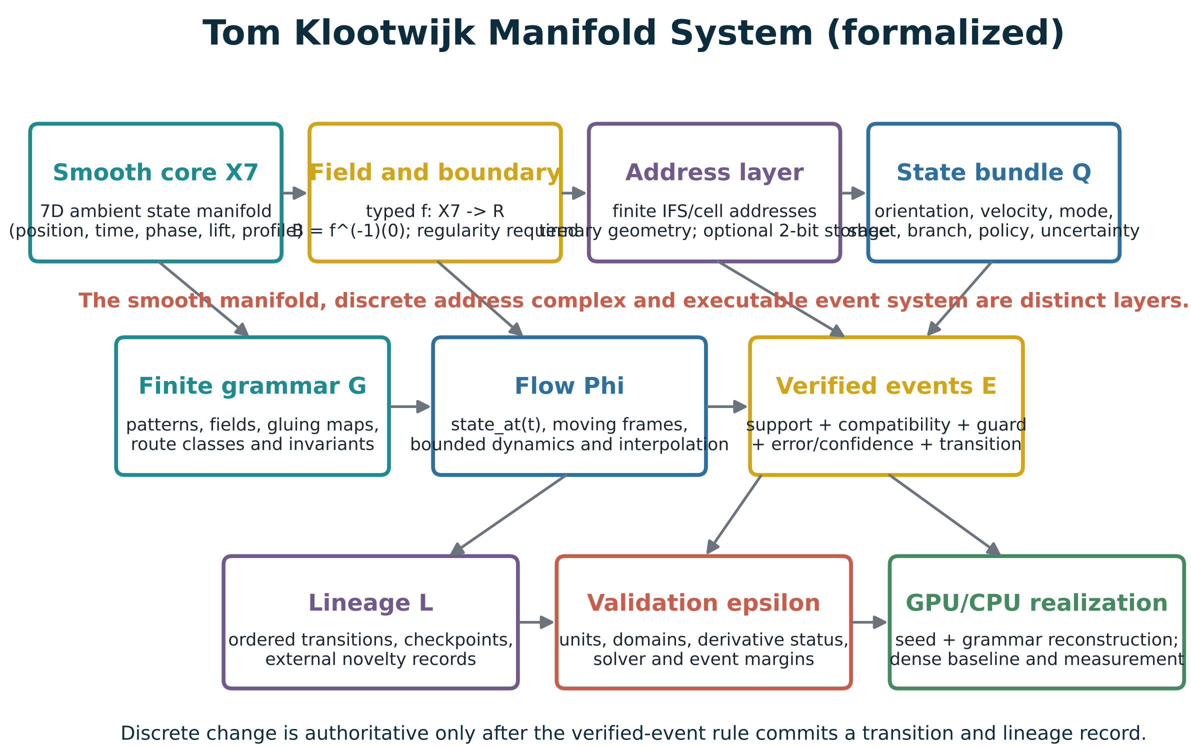


Figure 3.1: *Formalized architecture. The smooth state manifold, discrete address complex, field boundary, event calculus, and realization are intentionally separated.*

3.10 Axioms and conformance conditions

A conforming TKM implementation satisfies the following axioms.

Table 3.1: *TKM 1.0 axioms.*

Axiom	Requirement
A1	X_7 is explicitly declared as a seven-dimensional smooth ambient manifold with charts and a metric or metric contract.
A2	The discrete address complex $\mathcal{A}$ is typed separately from X_7; an address is not silently treated as a smooth coordinate.
A3	Coordinates are not authoritative identity. Sheet, route, branch, phase, and lineage can distinguish repeated coordinates.
A4	Every field declares whether it is exact signed distance or merely implicit, with units, domain, derivatives, and numerical error.
A5	Smooth-boundary status requires a regular-value check or a stated theorem that implies it. Singular points receive a status, not a false manifold label.
A6	Φ_t is closed form or numerically bounded. State queries include validity interval and residual/error.
A7	Discrete change is authorized only by support, compatibility, accepted guard status, confidence, and event margin.
A8	Every committed transition appends reconstructible lineage and any irreducible novelty.
A9	Address depth, grammar depth, event density, branch density, solver steps, and storage are bounded or trigger a kill criterion.
A10	A compact representation is promoted only after reconstruction at the declared queries is demonstrated within ε.
A11	Hardware claims are tied to a named target, compiler, precision, workload, timing method, power/thermal context, and baseline.
A12	Attribution metadata is separated from mathematical validity, executable integrity, and cryptographic provenance.

Chapter 4

Chronotemporal latch and smoothing, stated rigorously

4.1 Chronotemporal latch

The conversational phrase “chronotemporal latch” is useful if it is defined as a software invariant rather than as physical teleportation.

Definition 9: Tom Klootwijk chronotemporal latch

A TKM entity is **chronotemporally latched** over horizon H when:

1. state_at(t) is deterministic or returns a bounded uncertainty set for every declared $t \in H$;
2. reconstructing from the seed, grammar, checkpoints, lineage, and external inputs agrees with the authoritative state within $\varepsilon(t)$;
3. discrete state changes occur only at verified events and are totally ordered, commutatively merged, or explicitly recorded as conflicts;
4. repeated coordinates at different times, sheets, or routes remain distinct through lineage.

This definition captures the source desire for time-consistent procedural motion. It does not claim that time is encoded in one LSB, that omitted frames require zero computation, or that a bit flip physically changes the past.

4.2 A controlled rounding operator

Let $K_d \subset \mathbb{R}^m$ be a finite-depth approximation to a fractal or cell complex. Define

$$f_{r,d}(x) = d_{K_d}(x) - r.$$

The zero set $f_{r,d}^{-1}(0)$ is an offset shell at distance r. It rounds features smaller than the offset scale in a geometrically meaningful way, but it may still have nonsmooth points where nearest points are nonunique.

A smoother field may be formed by mollification:

$$\widetilde{f}_{\sigma,r,d}(x) = (\rho_\sigma * d_{K_d})(x) - r,$$

where ρ_σ is a smooth kernel. If 0 is a regular value, the zero set is smooth. This construction gives a rigorous version of the desire to surround a jagged finite-depth structure with a rounded shell.

Smoothing boundary

Mollification and smooth-min generally destroy exact-distance values and can change topology. Therefore r, σ, topology changes, and approximation error belong in the validation contract. “Rounder” is scale-dependent, not a free consequence of writing “SDF 0.”

4.3 Sphere tracing is not a one-step theorem

Sphere tracing advances a ray using a safe distance bound. Its advantage is that a valid bound avoids penetrating the surface. It still needs repeated field evaluations in general, and fractal or highly detailed fields may require many iterations. A zero-level condition identifies the hit surface; it does not analytically jump every ray to the solution.

A TKM renderer may use sphere tracing, interval methods, analytic roots, or rasterization. The rendering algorithm is downstream of the field contract. The formal system is not itself a renderer, matching the reading rule in UGTS-KC 2.0.

Chapter 5

Exact interpretation of the mnemonic notation

5.1 The source mnemonic

The namesake source proposes an informal notation of the form

$$\mathcal{TK}_{(10\text{-}07\text{-}1990)}(4,4,4,4) \Longrightarrow \mathrm{SDF}_0 \downarrow \lfloor 3 \rfloor_{\mathrm{LSB}}.$$

As written, this is not a complete mathematical sentence because the tuple components, field domain, arrow, down-selector, and LSB operation are not typed. TKM 1.0 retains it only as a mnemonic and assigns each token an exact role.

Table 5.1: *Formal interpretation of the mnemonic.*

Token	Admissible meaning	Meaning it does not acquire automatically
$\mathcal{TK}$	Namespace for this proposed specification and its versioned operators.	Proof of scientific priority, ownership, or external recognition.
$(4,4,4,4)$	A four-component parameter block θ whose fields must be named by schema.	A seven-dimensional manifold, a hypercube, a Sierpinski order, or a physical bounding volume without further definition.
SDF_0	Constraint $f_\theta(q) = 0$ for a declared field.	A sphere, smoothness, exact distance, or a one-step ray intersection.
$\lfloor 3 \rfloor$	Zero-based index $i = 3$, selecting the fourth component of a sequence with length at least four.	A dimension collapse, topology operation, or proof that seven directions meet.
LSB	A serialization/display convention or a bit-mask operation on a declared integer field.	A change in Hamming distance, information entropy, geometry, or time.

5.2 Canonical typed query

The rigorous replacement is

$$\mathrm{TKQuery}_i(q_0, t; \theta) = \mathrm{Project}_i\big(\Phi_t(q_0)\big) \quad \text{subject to} \quad f_\theta\big(\Phi_t(q_0)\big) = 0,$$

with $i = 3$ only when the projection target is declared and has at least four components. The function returns

$$(\mathsf{value}, \mathsf{status}, \mathsf{residual}, \mathsf{error}, \mathsf{lineage}).$$

A bit-level representation is a separate serialization function

$$\mathsf{encode}_{\mathrm{LSB}} : Q \to \{0,1\}^n.$$

The encoding may be compact, but the semantic query is not defined by the order in which bits are printed.

5.3 The 943 to 937 calculation

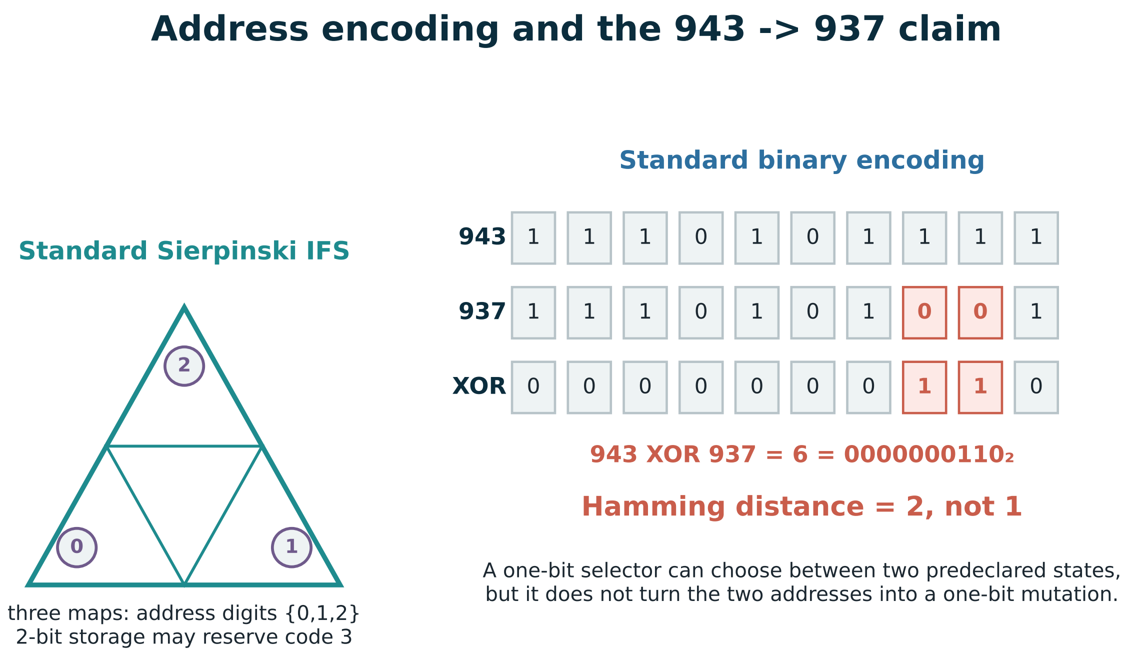


Figure 5.1: *The standard Sierpinski address alphabet and the exact binary comparison of 943 and 937.*

A legitimate one-bit *selector* construction is possible:

$$\mathsf{selectedState}(b) = \begin{cases} 943, & b = 0, \\ 937, & b = 1. \end{cases}$$

Flipping b changes which already-declared state is selected. This is not a one-bit mutation of the ten-bit representation of 943 into the ten-bit representation of 937. It shifts complexity into the decoder, lookup, grammar, or two stored/reconstructible alternatives. This distinction is central to honest compression claims.

Claim-by-claim mathematical audit

6.1 Audit summary

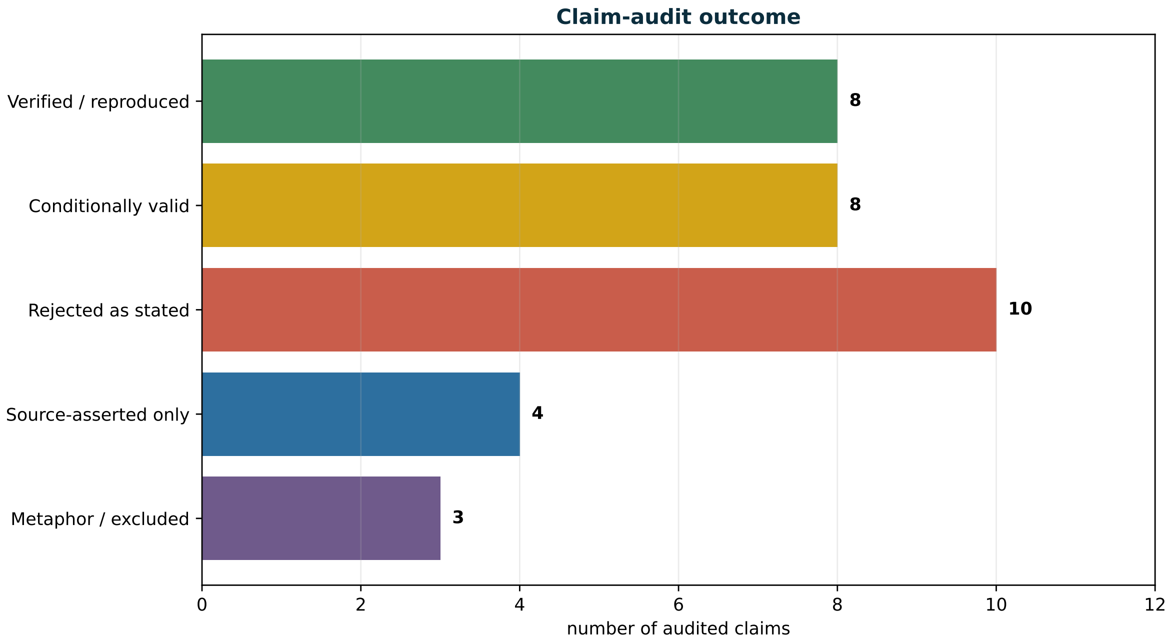


Figure 6.1: *Outcome counts for the 33 audited claims listed in table 6.1.*

The audit does not treat rejection as failure of the whole project. Several rejected statements are exactly the kind of overreach that S1 and S4 already warn against. Removing them makes the remaining system more precise and testable.

Table 6.1: *Full claim audit.*

ID	Claim or motif	Status	Finding
V1	Seven mutually perpendicular axes can exist in a seven-dimensional inner-product space.	**VERIFIED**	True by linear independence of nonzero mutually orthogonal vectors. Seven axes require dimension at least seven.
V2	$943 = 1110101111_2 = 32233_4$ and $937 = 1110101001_2 = 32221_4$.	**VERIFIED**	Exact integer conversion reproduced by an independent script.
V3	The transition $943 \to 937$ differs in two binary positions.	**VERIFIED**	XOR is $6 = 0000000110_2$ and popcount is two.
V4	In the usual sign convention, negative/positive/zero field values can represent inside/outside/boundary.	**VERIFIED**	Correct when the field and orientation convention are declared. Reversing the sign convention is also possible.
V5	The standard Sierpinski triangle has three contraction maps and similarity dimension $\log 3/\log 2$.	**VERIFIED**	Standard IFS calculation; the numerical value 1.5849625007 was independently recomputed.
V6	The supplied package manifest is internally intact.	**VERIFIED**	82 of 82 SHA-256 manifest entries matched; zero failures.
V7	The CPU implementation builds and its included tests pass.	**VERIFIED**	Clean GCC 14.2/C++20 build with CUDA disabled; two of two CTest suites passed.
V8	The bundled seven-day CPU pass-event CSV is reproducible.	**VERIFIED**	The full command regenerated all counts and produced a bit-for-bit identical CSV with the same SHA-256.
C1	A "Tom Klootwijk manifold" can be formally defined as a seven-dimensional ambient state manifold.	**CONDITIONAL**	Valid as a proposed definition when charts, smoothness, and a dimension contract are supplied. The name itself does not establish novelty or priority.
C2	$f^{-1}(0)$ is a smooth boundary manifold.	**CONDITIONAL**	True if f is smooth and zero is a regular value. In a seven-dimensional ambient manifold the regular boundary has dimension six.
C3	A field is an exact signed-distance field.	**CONDITIONAL**	True only when it equals signed metric distance on the declared domain. Many implicit fields, blends, and smoothed fields are not exact SDFs.
C4	A fractal can be given a rounded smooth shell.	**CONDITIONAL**	A finite-depth set can be offset and mollified; a regular zero value yields a smooth shell. Scale, topology change, and approximation error must be declared.
C5	A compact seed/grammar can replace dense stored states.	**CONDITIONAL**	Valid for reconstructible families at stated queries and error. External novelty, high event density, or repeated reuse can remove the advantage.
C6	One bit can switch between 943 and 937.	**CONDITIONAL**	A one-bit selector can choose between two predeclared states. It does not mutate one ten-bit value into the other or reduce their Hamming distance.
C7	A chronotemporal latch can avoid materializing every display frame.	**CONDITIONAL**	Valid as deterministic $\mathsf{state\_at}(t)$ plus checkpoint/lineage reconstruction. It is not physical time travel and may still require substantial arithmetic.
C8	Direct seed queries can beat dense materialize-plus-query.	**CONDITIONAL**	Bundled RTX results show a narrow 0.93–2.50% p50 advantage for the tested one-off workloads, while resident dense queries remain roughly 4.5x faster.
R1	The four-tuple $(4,4,4,4)$ supplies seven mutually orthogonal axes.	**REJECTED**	Four coordinates span at most four independent directions unless another seven-dimensional space is explicitly defined.
R2	$(4,4,4,4)$ automatically denotes a four-dimensional hypercube or bounding volume.	**REJECTED**	A point/tuple is not a hypercube. A hypercube needs a set of inequalities, center/extent convention, or vertex/half-space description.

Continued on next page

ID	Claim or motif	Status	Finding
R3	Writing "SDF 0" automatically creates a sphere or hypersphere.	**REJECTED**	Every regular implicit surface is a zero set of some field; the geometry depends on f, not on the number zero.
R4	A standard Sierpinski triangle is intrinsically base four.	**REJECTED**	It has three self-similar branches. Quaternary packing is an optional code with a reserved state, not a topological fact.
R5	$943 \longrightarrow 937$ is a single-bit mutation.	**REJECTED**	Hamming distance equals two.
R6	Zero-based indexing or LSB-first notation changes the Hamming distance.	**REJECTED**	These are labeling/display conventions; applying the same reindexing to both strings preserves unequal-coordinate count.
R7	A parity flip teleports a full state through space or time.	**REJECTED**	Parity is an integrity/coding mechanism. State selection or reconstruction requires a declared decoder and retained information.
R8	The procedural representation has a literal zero-byte VRAM footprint.	**REJECTED**	Instructions, parameters, registers, seed buffers, outputs, runtime state, and driver allocations consume storage even when no dense mesh/timeline is resident.
R9	The notation proves universal $O(1)$ rendering or a single-cycle arbitrary query.	**REJECTED**	Complexity depends on input size, grammar depth, precision, field evaluations, root/event solving, output size, and hardware.
R10	The construction guarantees infinite detail, perfect smoothness, no aliasing, and flawless temporal stability.	**REJECTED**	Finite precision, sampling, solver tolerances, discontinuities, and hardware limits remain. Such guarantees need a much narrower theorem.
S1	The RTX 5070 Ti Laptop p50 timing tables are accurate.	**SOURCE-ASSERTED**	The CSVs and logs are internally consistent and identify compute capability 12.0, but no compatible GPU was available for an independent rerun.
S2	Direct and dense compact GPU event sets match at 717, 1,243, and 4,970 events.	**SOURCE-ASSERTED**	Bundled reports/CSVs say sorted payloads matched with no truncation. Not independently rerun here.
S3	The bundled Nsight, power, and thermal measurements characterize the target laptop.	**SOURCE-ASSERTED**	Plausible and internally documented, but target-specific and not independently measured.
S4	A full-horizon GPU oracle has a one-count support mismatch while verified event count remains 717.	**SOURCE-ASSERTED**	Reported by the supplied laptop-verification document; the failing boundary was not reproduced without the target GPU.
M1	Oppenheimer's "latter" statement mathematically designates the sign of a graphics SDF.	**METAPHOR**	Historical/political language does not define a graphics field. The analogy is not used as evidence.
M2	Hadamard $\|+\rangle/\|-\rangle$ states are the same object as SDF outside/inside signs.	**METAPHOR**	Both use plus/minus notation, but one is a quantum basis and the other is a scalar sign convention. No physical identification follows.
M3	Square-and-compasses symbolism proves the geometry.	**METAPHOR**	It can be an artistic explanation of orthogonality versus roundness, but it supplies no axiom, invariant, or validation result.

Chapter 7

Independent verification of the KLB package

7.1 Scope of independent work

The ZIP was extracted into an isolated working directory. Verification was performed without modifying source code. The independently executed scope was:

- SHA-256 manifest verification;
- CPU-only CMake configuration and clean build;
- CTest execution;
- exact comparison of `inspect` and `verify` outputs against supplied captures;
- full seven-day pass-event regeneration;
- exact byte/hash comparison of the regenerated event CSV;
- independent integer, fractal-dimension, storage-ratio, and timing-ratio calculations.

CUDA execution, Nsight profiling, laptop power/thermal measurements, and the stronger full-horizon GPU oracle were not independently rerun because the current environment had no compatible NVIDIA GPU.

7.2 Integrity and build results

Table 7.1: *Independent package verification results.*

Check	Result	Evidence
Manifest	PASS	82 entries OK; zero failed.
CMake configure	PASS	Release, CUDA disabled, GCC 14.2, C++20.
Build	PASS	CPU applications and test targets compiled cleanly.
CTest	PASS	100% tests passed; 0 failed out of 2.
KLOC inspect	PASS	Exact text match to bundled capture; zero diff lines.
KLOC verify	PASS	Exact text match to bundled capture; zero diff lines.
Pass schedule	PASS	Recomputed CSV has zero diff lines and the same SHA-256 as the bundled file.

Reproduced seven-day counts

Quantity	Reproduced value
One-second intervals	604,800
Satellite-time candidates	19,353,600
Support survivors	19,214,155
Compatible survivors	19,214,155
Visible sample states	5,498,030
Acquisition/loss events	717

The regenerated CSV hash is

`a08b9295d29c94dac7244d3a70aea2e0196ad56e5c25733c909e4fbdf68d4f86`.

This establishes implementation determinism for the tested CPU build and command. It does not establish that the orbital states are accurate relative to SGP4 or precise ephemerides.

7.3 What the KLB package operationalizes

KLB deliberately turns broad motifs into narrower mechanisms:

- a 37-bit log-spherical point record with an explicit parity bit;
- a finite parity-controlled grammar;
- a discrete Klein-style quotient for legal in-buffer routing, not physical VRAM topology;
- seed/predictor reconstruction with checkpoints and novelty;
- local support, route compatibility, signed-field guard, and compact verified events;
- a KLOC1 orbit-seed container with 32 mean-element seeds and seven hash-linked daily nodes.

That narrowing is a strength. The package explicitly rejects claims that it forces physical memory into a Klein bottle, eliminates allocation boundaries, proves arbitrary compression ratios, or establishes speculative hardware behavior.

7.4 Model-based expansion ratio

The 3,809-byte KLOC1 container represents 32 seeds and seven timeline nodes used to reconstruct a declared seven-day, one-second state horizon. The equivalent dense float4 positions under the same bundled predictor occupy 309,658,112 bytes, giving

$$\frac{309{,}658{,}112}{3{,}809} \approx 81{,}296.4327.$$

This is a *horizon-relative model expansion ratio*, not lossless compression of a pre-existing 295.3 MiB trajectory file. Both the compact and dense benchmark paths generate/query the same predictor, which makes the workload comparison meaningful, but the ratio cannot be generalized to arbitrary motion.

Bundled GPU evidence and performance interpretation

8.1 Measured p50 latencies

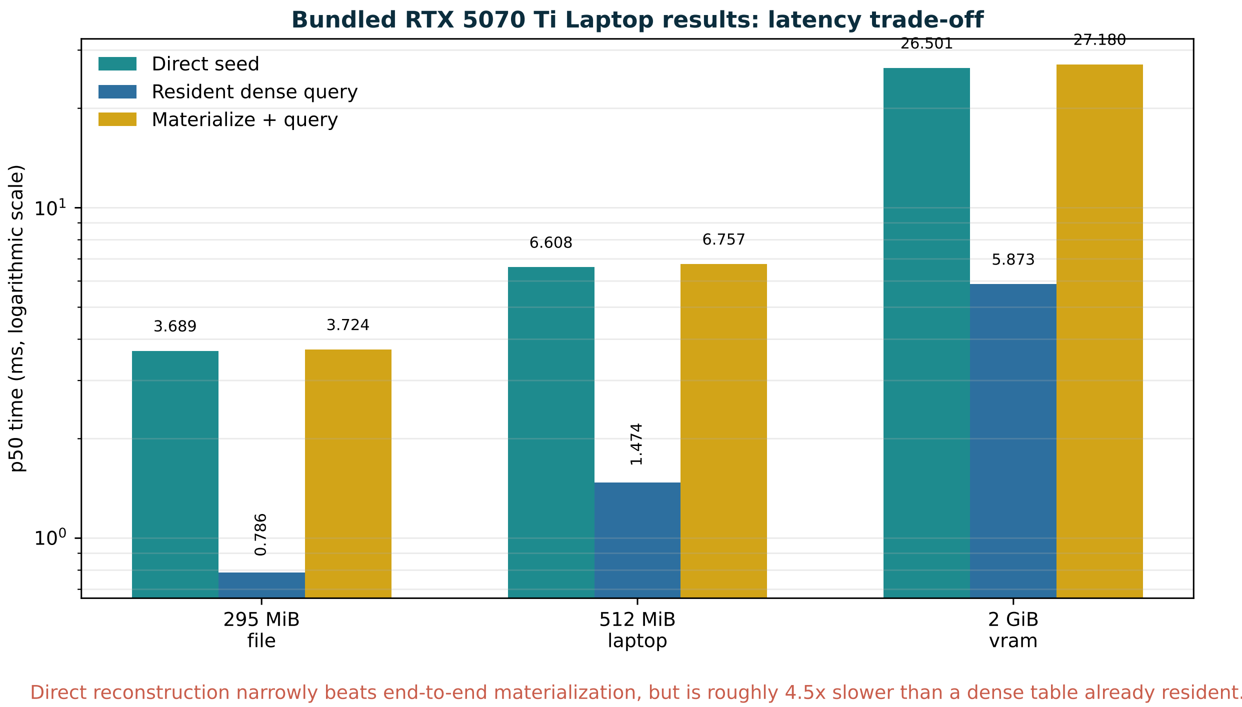


Figure 8.1: *Bundled RTX 5070 Ti Laptop p50 latencies. The direct path narrowly beats end-to-end dense but not an already-resident dense table.*

Table 8.1: *Bundled GPU timing evidence (source-asserted; p50 milliseconds).*

Preset	Direct seed	Materialize	Resident dense	End-to-end dense	Direct compact	Dense compact
295.3 MiB file	3.689	3.062	0.786	3.724	4.287	1.290
512 MiB laptop	6.608	5.313	1.474	6.757	7.496	2.441
2 GiB VRAM	26.501	21.356	5.873	27.180	30.013	9.807

The proper comparison depends on the question:

- For a one-off query where the dense horizon would otherwise have to be materialized, direct seed reconstruction wins narrowly in the supplied p50 measurements.
- For repeated queries when the dense table is already resident, dense lookup is roughly 4.5 times faster.

- Compact event output adds measurable overhead even though event yield is very low.

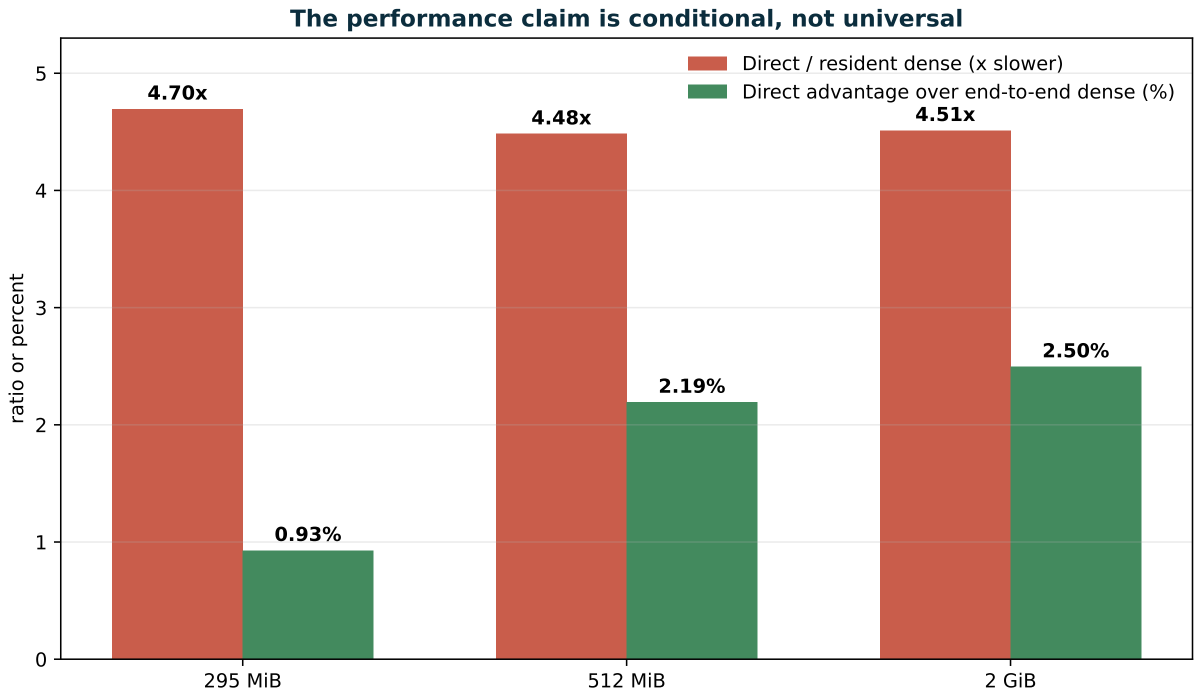


Figure 8.2: *The performance claim is conditional: a small end-to-end advantage coexists with a large resident-dense disadvantage.*

8.2 Exact crossover calculations

From the bundled p50 values:

$$\begin{aligned}
\text{file advantage} &= 100\frac{3.723594 - 3.688999}{3.723594} = 0.929\%,\\
\text{laptop advantage} &= 100\frac{6.756591 - 6.608373}{6.756591} = 2.194\%,\\
\text{VRAM advantage} &= 100\frac{27.179630 - 26.501227}{27.179630} = 2.496\%.
\end{aligned}$$

The file direct path is

$$\frac{3.688999}{0.785619} \approx 4.696$$

times slower than resident dense lookup. Similar ratios are 4.483 and 4.512 for the larger presets.

The correct conclusion is therefore *crossover under a specific reuse and memory-materialization model*, not "procedural is always faster."

8.3 Correctness boundary in the bundled laptop report

The bundled laptop-verification report states that the required 4,096-epoch CPU/GPU oracle prefix matched, direct/dense counters matched in the three primary runs, compact event payloads matched at 717, 1,243, and 4,970 events, and no output was truncated. It also reports a stronger full-horizon check in which CPU support/compatibility count was 19,214,155 and GPU count was 19,214,154, while verified event count remained 717.

This one-count discrepancy is small in effect but important in evidence status. It means "complete full-horizon exact CPU/GPU counters" is not established. The likely class of issue is a floating-point guard/support boundary, but the root candidate must be isolated before promotion.

8.4 Orbital-accuracy boundary

The package uses a deterministic Kepler solve with pack-time secular J2 rates. CelesTrak GP elements are fitted for use with SGP4. The package itself correctly states that the current deployment is for compression/performance work and coarse scheduling, not navigation, collision avoidance, safety of flight, or precise antenna pointing.

A useful operational promotion would require at least:

- an SGP4 reference trajectory for every bundled object and epoch sample policy;
- position/velocity error distributions and worst cases;
- proof that guard/event ordering remains within the declared elevation/time margin;
- cross-compiler and CPU/GPU tolerance policies for boundary candidates.

8.5 Provenance/versioning issue

The package's `docs/VALIDATION.md` describes GPU compilation and execution as not completed in the preparation environment, while later root-level logs, CSVs, Nsight reports, telemetry, and the laptop-verification DOCX document completed target-laptop runs. This is not necessarily contradictory chronology, but the package should version-stamp the validation stages so readers can distinguish preparation-stage evidence from later deployment evidence.

Chapter 9

Mapping to UGTS-KC 2.0 and KLB

9.1 TKM as a specialization, not a replacement

UGTS-KC 2.0 defines an expanded substrate tuple

$$(G, P, F, K, D, R, S, C, T, I, L),$$

covering grammar, patterns, fields, kinematics, dynamics, guards, support, compatibility, transitions, topology invariants, and lineage. TKM 1.0 is a named specialization that adds an explicit seven-dimensional ambient manifold and an explicit regular-boundary rule.

Table 9.1: *Mapping from TKM 1.0 to UGTS-KC 2.0.*

TKM component	UGTS-KC component(s)	Relationship
$(X_7, g), Q$	typed state	Makes the ambient dimension and attached state bundle explicit.
f	F, R	Field plus relation/guard; exact-SDF status is typed.
$\mathcal{A}, \pi$	I plus lineage/route fields	Address, covering, winding, cell, and chart mapping.
Γ	G, P, F, I	Finite grammar emitting typed patterns, fields, and topology descriptors.
Φ	K, D	Closed-form or bounded kinematic/dynamic state evolution.
$\mathcal{E}$	R, S, C, T	Support, compatibility, guard classification, verified transition.
$\mathcal{L}$	L	Ordered lineage, novelty, checkpoints, and conflict records.
ε	validation contracts and kill criteria	Centralizes units, bounds, residuals, event margin, and performance gates.

The UGTS-KC warning that coordinates remain non-authoritative for identity is especially important for self-intersecting curves, quotient charts, periodic routes, and seed reconstruction. TKM adopts it as Axiom A3.

9.2 Mapping to KLB operational mechanisms

Table 9.3: *KLB package as a partial TKM realization.*

TKM layer	KLB realization	Boundary
State/address	37-bit log-spherical records, point indices, KLSC/KLOC nodes, finite symbols, parity	A packed record/address scheme, not a proof of a smooth seven-dimensional manifold.
Topology/route	Discrete Klein quotient, route bit, parent links, node order	Legal combinatorial routing in allocated buffers, not physical non-orientable VRAM.
Grammar/flow	Parity grammar, predictors, cone/-sphere fields, orbit seed propagation	Reconstructible only for declared model families and error contracts.
Support/compatibility	Radial/angular or station visibility support plus route compatibility	Candidate pruning, not an interaction by itself.
Guard/event	Sphere/elevation signed guard and compact append	Closest operational match to the TKM verified-event pipeline.
Lineage/novelty	Hash-linked nodes, checkpoints, sparse residuals, event records	FNV hashes are integrity checks, not cryptographic authorship proof.
Validation	Tests, exact counters, benchmark CSVs, profiler logs, kill criteria	CPU behavior reproduced; GPU and physical accuracy remain target-specific.

Chapter 10

Reference API, schema, and conformance tests

10.1 Minimal machine-readable schema

A conforming implementation should expose enough metadata that the mathematical object and the execution are not confused. The following JSON-shaped example is normative in structure but illustrative in values.

Listing 10.1: *TKM 1.0 example definition.*

```json
{
  "schema": "tkm-1.0",
  "attribution": {
    "name": "Tom Klootwijk",
    "requester_supplied_identifier": "NL200678942",
    "requester_supplied_date_of_birth": "1990-07-10",
    "verification_status": "requester-supplied-not-independently-verified"
  },
  "ambient": {
    "dimension": 7,
    "model": "R3_position x R_time x S1_phase x R_route_lift x R_profile",
    "metric": "declared-riemannian-metric-v1"
  },
  "state_bundle": {
    "orientation": "SO(3)",
    "fields": ["velocity", "angular_velocity", "mode", "sheet",
               "branch", "policy", "uncertainty", "lineage"]
  },
  "field": {
    "kind": "implicit-or-exact-sdf",
    "formula_id": "versioned-operator-id",
    "zero_level": 0.0,
    "regularity_required": true,
    "units": "declared",
    "error_budget": "epsilon_f"
  },
  "address": {
    "alphabet": [0, 1, 2],
    "storage_bits_per_symbol": 2,
    "reserved_codes": [3],
    "max_depth": 16
  },
  "flow": {"family": "closed_form_or_bounded_integrator"},
  "event": {
    "requires": ["support", "compatibility", "accepted_guard_status",
                 "confidence_floor", "numeric_error_below_margin"]
  },
  "lineage": {"authoritative_transitions": true, "external_novelty": true}
}
```

10.2 Core query interface

Table 10.1: *Reference TKM API.*

Function	Returns	Contract
`state_at(entity,t)`	typed state + status	Closed form or bounded integration; includes validity interval and error.
`field_at(field,q)`	scalar + derivatives	Distinguishes exact SDF from implicit field.
`frame_at(pattern,s)`	frame + degeneracy	Frenet, Bishop, or body policy; no invented frame at singularity.
`route_class(path)`	address/topology descriptor	Winding, covering, braid, cell, or chart data as declared.
`next_event(entity,t0,t1)`	earliest verified event	Support and compatibility precede guard resolution.
`reconstruct(entity,t)`	state + lineage proof	Seed/grammar/checkpoint/external novelty reconstruction within ε.
`round_field(K,r,sigma)`	implicit field + status	Offset/mollified field; does not claim exact distance unless verified.
`encode_lsb(state)`	bit string + schema hash	Serialization only; semantic identity and Hamming claims remain explicit.

10.3 Verified-event pseudocode

Listing 10.2: *Reference verified-event logic.*

```
def verify_event(entity, candidate, policy):
    q = state_at(entity, candidate.time)
    if not candidate.support.contains(q):
        return Rejected("outside_support")
    if not candidate.compatibility.accepts(q):
        return Rejected("incompatible")

    guard = candidate.guard.evaluate_with_interval(q)
    status = classify_degeneracy(guard, policy)
    if status not in policy.accepted_statuses:
        return Rejected(status)
    if guard.confidence < policy.confidence_floor:
        return Rejected("low_confidence")
    if guard.numeric_error > policy.event_margin:
        return Rejected("error_exceeds_margin")

    patch = candidate.transition.apply(q)
    lineage = append_authoritative_event(q, patch, guard, status)
    return Verified(patch, lineage, guard)
```

10.4 Conformance test suite

A minimal TKM conformance suite should include:

1. **Dimension test:** every chart of X_7 maps to an open subset of $\mathbb{R}^7$ with compatible transitions.

2. **Regular-boundary test:** accepted smooth boundary samples satisfy $\|df\| > \delta$ or are covered by a stronger regularity proof.
3. **Field-kind test:** exact SDF claims verify distance/eikonal behavior within a declared domain; implicit fields are not mislabeled.
4. **Address test:** all symbols, reserved codes, depth bounds, and chart maps are validated.
5. **Bit invariant test:** encoding conversions and Hamming distances use exact integer arithmetic.
6. **Event test:** unsupported or incompatible zero crossings do not commit transitions.
7. **Degeneracy test:** crossing, touch, tangency, coincidence, and simultaneous events receive declared policies.
8. **Lineage test:** split/merge ancestry and external novelty reconstruct the authoritative state.
9. **Numerical kill test:** singular parameters, unstable frames, or excessive residuals stop rather than silently continue.
10. **Performance test:** direct, resident-dense, materialization, and end-to-end baselines are reported separately at equal error.

Chapter 11

Falsifiability, kill criteria, and next verification gates

11.1 Mathematical kill criteria

Reject a claimed TKM manifold/boundary instance when:

- the ambient dimension is inferred from notation rather than declared by charts;
- a supposed smooth zero set contains unhandled critical points $df = 0$;
- a generic implicit field is advertised as exact distance without verification;
- a discrete address graph is called a smooth manifold without a chart construction;
- coordinates are used as identity after self-intersection, quotient wrapping, or route re-entry;
- topology edits violate incidence, orientation, or chart-cocycle invariants.

11.2 Numerical and event kill criteria

Stop or downgrade a query when:

- field/derivative residual exceeds the event margin;
- speed, curvature, frame normalization, or spline denominator falls below a declared numerical floor;
- equal-time intervals cannot be ordered or merged without conflicting authoritative patches;
- chatter or Zeno-like accumulation exceeds event-density/storage bounds;
- reconstructibility fails or compact checksums collide within the application budget.

11.3 Performance kill criteria

Reject a performance promotion when:

- direct reconstruction is slower than the avoided materialization/storage cost at equal error for the target query pattern;
- the workload repeatedly reuses a resident dense table and the compact path cannot justify its slower queries;
- grammar divergence, transcendental work, root solving, register pressure, or event compaction dominates;
- novelty becomes dense or checkpoints remove the expected storage benefit;
- a conventional dense, indexed, or physics-specific implementation is cheaper, faster, or more accurate.

11.4 Evidence needed for stronger claims

Table 11.1: *Promotion gates for claims beyond TKM 1.0.*

Desired stronger claim	Required evidence
Recognized new mathematical object	Independent expert review, clear relationship to existing manifold/level-set/cell-complex literature, public versioned specification, and reproducible examples.
Exact SDF	Proof or numerical certification of signed metric distance on the declared domain, including medial-axis/regularity boundaries.
Smooth rounded fractal	A finite-depth/offset/mollification definition, regular-value proof, topology analysis as r, σ vary, and error against target geometry.
One-bit state transition	A precise coding/decoder theorem specifying what information is pre-stored or generated; no conflation with direct mutation Hamming distance.
Universal $O(1)$ query	A fixed input-size model or a theorem that bounds grammar depth, precision, event output, and memory access independently of the variable being scaled.
GPU superiority	Independent reruns on named GPUs, compiler/toolchain capture, equal-error baselines, confidence intervals, thermal/power context, and query-reuse analysis.
Operational orbit scheduling	SGP4/precise reference comparison, event-order/error budget, long-horizon validation, and resolution of the full-horizon CPU/GPU boundary mismatch.
Authorship/priority	Independent documentary or legal evidence; mathematical validity and package hashes are insufficient.

Final canonical definition and verdict

12.1 Canonical definition

Final formal definition

The **Tom Klootwijk Manifold System** is the structured tuple

$$\mathbf{TKM}_1 = \big(X_7, g;\ Q;\ f;\ \mathcal{A}, \pi, \Gamma;\ \Phi;\ \mathcal{E};\ \mathcal{L};\ \varepsilon\big),$$

where:

- (X_7, g) is a seven-dimensional smooth Riemannian ambient state manifold;
- $Q \to X_7$ is a typed state bundle carrying orientation, motion, discrete mode/sheet/branch, uncertainty, and lineage;
- $f : X_7 \to \mathbb{R}$ is a typed exact-SDF or implicit-field operator with a derivative and error contract;
- $B_{\mathrm{TK}} = f^{-1}(0)$ is the boundary, a smooth six-dimensional submanifold when zero is a regular value;
- $\mathcal{A}$ is a bounded discrete address complex, π its chart/support map, and Γ a finite typed grammar;
- Φ supplies closed-form or bounded kinematic/dynamic state-at-time queries;
- $\mathcal{E}$ admits discrete change only after support, compatibility, accepted guard status, confidence, and numerical margin are satisfied;
- $\mathcal{L}$ stores authoritative lineage, checkpoints, conflicts, and irreducible external novelty;
- ε states units, bounds, residuals, event margins, determinism requirements, and kill criteria.

A compact representation is valid only for declared reconstructible families and queries. Bit indexing is serialization, not topology. Hardware performance and physical accuracy remain separate measured claims.

12.2 Verdict on the requested definition

Overall verdict

GO for the formal TKM 1.0 definition above as a coherent specialization of the supplied UGTS query-first substrate.
GO for the independently reproduced KLB CPU integrity, build, tests, and seven-day deterministic pass schedule.
CONDITIONAL for the bundled GPU crossover result: it is a narrow one-off-query win against materialize-plus-query, not a win against resident dense lookup and not a universal rendering theorem.
NO-GO for the claims that zero-based indexing changes Hamming distance, that 943 → 937 is one data-bit mutation, that "SDF 0" automatically creates a sphere/smooth manifold, that VRAM use is literally zero, or that the notation proves universal $O(1)$ or single-cycle rendering.
NOT ESTABLISHED for legal authorship/priority, community recognition of the name, SGP4-quality orbit accuracy, navigation-grade use, and complete full-horizon CPU/GPU counter identity.

12.3 Plain-language summary

The strongest version of the idea is not "one magic notation makes impossible geometry free." It is this:

> Define a seven-coordinate state space. Keep its smooth geometry separate from its discrete address grammar. Use a signed or implicit field to state a boundary. Compute motion from a bounded rule instead of storing every frame when reconstruction is valid. Allow changes only when local support, compatibility, and a numerically verified guard agree. Preserve lineage so repeated positions do not erase identity. Measure the actual hardware trade-off instead of assuming compact notation is automatically faster.

That formulation is mathematically coherent, executable, falsifiable, and faithful to the best parts of the supplied substrate work.

Source hashes and provenance

Table 1: *SHA-256 of the four requester-supplied inputs.*

Input	SHA-256
UGTS-KC 2.0 PDF	`a29dd9959ebd300c32d0e1df8fa2c0d4ee15f428892557b98b5f18cd3e878399`
SDF/Oppenheimer conversational PDF	`b3ab8e1d9f47171efa4f8370c94f3b4a48b58eb5bf30970312c7573d249b6aa3`
Namesake manifold conversational PDF	`5004ef709ddd2de3350069642fdd7c78f4cdf6aeea15a2e703b952766f6b97d2`
KLB SeedChain GPU 0.3.0 ZIP	`c3e00e94ff1c3c13ac9ae76ea84a95117551e022375cefda8c706a7bb14d1ad4`

The package's authorship notice says the requester attribution is reproduced as user-supplied provenance and is not an independent legal determination. TKM 1.0 follows the same boundary.

Reproduction commands and captured outcomes

.1 Manifest and CPU build

```
sha256sum -c MANIFEST.sha256

cmake -S . -B build-cpu \
  -DCMAKE_BUILD_TYPE=Release \
  -DKLB_BUILD_CUDA=OFF
cmake --build build-cpu -j
ctest --test-dir build-cpu --output-on-failure
```

Captured summary:

```
manifest_ok=82
manifest_failed=0
ctest=100% tests passed, 0 tests failed out of 2
inspect_diff_lines=0
verify_diff_lines=0
pass_diff_lines=0
```

.2 Full pass-event reproduction

```
./build-cpu/klb_orbit passes \
  data/orbit/gps_ops_2026-08-16_7d_1s.kloc \
  --lat 52 --lon 5 --alt-km 0.05 \
  --elevation-deg 10 --crossing-band-deg 0.25 \
  --hours 168 --step-seconds 1 \
  --output pass_events_recomputed.csv
```

The recomputed file and bundled file were byte-identical in this environment. The application counts are listed in table 7.1.

.3 Independent mathematical checks

```
def base4(n):
    digits = []
    while n:
        n, r = divmod(n, 4)
        digits.append(str(r))
    return ''.join(reversed(digits or ['0']))

hamming_943_937 = (943 ^ 937).bit_count()  # 2
sierpinski_dim = math.log(3) / math.log(2)
working_set_ratio = 309_658_112 / 3_809
```

Bibliographic and technical references

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